



CHALLENGE PROJECT PAP

Modified Stripline Design and Implementation of III-V Semiconductor MMICs

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December 10, 2025

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1 Executive Summary

1.1 Abstract

The purpose of this proposal is to demonstrate the need for vertically integrated monolithic microwave integrated circuits for wide-band phased array applications. The proposal will meet all requirements of the Gordon Institute of Engineering Leadership at Northeastern University for the required Challenge Project. This Challenge Project must include both technical and leadership aspects in order to provide realized change within the sponsor company, RTX - Raytheon. This proposal will focus on the development of integrated circuits which are crucial for phased array radar operation and can dictate the function of a radar. While this is a comprehensive subject and requires multiple teams to develop a wafer process within Raytheon, the candidate will take the perspective of a MMIC designer who will consider the radio frequency effects within the process as to characterize the performance compared to traditional legacy design processes. The candidate will do this by investigating the effects of differing topologies for active components such as transistors and passive components such as matching networks which are made up of resistors, capacitors, inductors, and transmission lines. By doing this, the candidate will offer Raytheon a viable way for designers to develop millimeter wave products at a lower cost which will have a significant impact on the business and any organization that utilizes Raytheon's radar systems.

1.2 Project Mission Statement

This project intends to systematically assess the feasibility of designing integrated circuits in a way that allows for Raytheon to fully integrate a new process so that the cost to Raytheon is reduced, while providing a technical solution for the increasing need for remote sensing systems such as phased array RADAR. This need for remote sensing solutions is derived from the desire to provide the U.S. Military and its allies with the tools to help better protect themselves against foreign threats. This project offers a step toward the future of integrated circuit development so that RADAR systems can be provided that enhance military capabilities while reducing the cost to manufacture and build such systems. Raytheon's mission will foremost and always be the protection of the U.S. Military personnel who serve our country.

1.3 Key Stakeholders

- In today's day and age, federal defense budget spending is staying constant or declining while the need for remote sensing of airspace for military applications is growing.
- This project aims to solve this by reducing the cost and improving the capabilities of major RADAR systems by changing the way microwave electronics are designed and fabricated through a novel process.
- The impact of this project propagates through the following entities:
 1. RTX - Raytheon
 2. The U.S. Military
 3. Any country who uses these RADAR systems to help protect their land and people
- The impact on these RADAR systems can be seen most recently in the Ukraine war where Patriot RADAR systems have been instrumental in the neutralization of enemy attacks on military and civilian assets, articles [1] and [2].

1.4 GIEL Cadidate Background

This project's candidate, Russell Anderson, received his B.S. degree in Electrical Engineering from the University of New Hampshire in 2020. The candidate then started his engineering career as an RF Test Engineer at Raytheon in 2020. While in this role, the candidate characterized monolithic microwave integrated circuits such as power amplifiers, low-noise amplifiers, switches, limiters, and other active and passive components within the RF engineering field. During his time, the candidate performed duties as the applications and test engineer in the Raytheon University Research Institute (RURI) at the University of Massachusetts Lowell where research programs applied additive manufacturing techniques to printable RF circuits. In addition to this, the candidate pursued his M.S. degree in Electrical Engineering from the University of Massachusetts Amherst with a focus in RF and microwave engineering. In 2022, the candidate transitioned to a MMIC design role where he contributes to the design, simulation, and layout of GaN and GaAs microwave integrated circuits.

1.5 Industry Background

The Advanced Microelectronic Solutions (AMS) department is a subsection of the Electrical and Digital Products Directorate within Raytheon, a business unit of RTX. Within this department lies various sections, all which work together in order to design, manufacture, and test monolithic microwave integrated circuits (MMICs) as well as transmit-receive (T/R) modules. AMS does this by operating a wafer foundry that produces III-V semiconductor devices, primarily with gallium nitride (GaN) and gallium arsenide (GaAs). These devices operate at radio and millimeter-wave (mmWave) frequencies that serve specific functions within various military applications which are purchased by the Department of Defense (DoD) for the United States Military.

These devices may include circuits amplifiers, switches, phase shifters, filters, etc., which are first conceptualized when either an internal or external customer creates a need for such devices that aren't commercially available. The circuits are designed using CAD software that can simulate the electrical performance which usually consists of comparing prior results from already fabricated and characterized circuits in the form of a backfit. Backfitting the CAD simulations is a necessary task as the wafer process itself has inherent variation within it. Variation in the wafer process can be quantified and is constantly being reported on and optimized for best results or yield. The design process can take anywhere from two to four months depending on the complexity and novelty of the design. Throughout this process various design reviews take place in order to ensure the robustness of the design and is done so by independent peer-reviewers. Within these reviews are members of each section in the department to ensure the design meets certain design rules and regulations for the foundry's processes as well as the initial testing and characterization.

Once the circuit design is complete, a team of layout and process engineers do a final review of the wafer layout and form the reticle which is a larger footprint that contains repeating structures of the chip along with any chip variations and test structures that may want to be explored. The design and layout of the new chip are complete and the six month foundry process begins as the wafer goes from a simple pure material to a complex finished product. Now the wafer's journey still continues as it goes through tests and measurements along the way and ends finally at the production test stage where its full electrical performance is tested and verified.

2 Project Technical Background

2.1 Phased Array Architectures and T/R Modules

RADAR or Radio Detection and Ranging is the use of RF/microwave frequency pulses to determine the position and track stationary or moving targets. In order to do this a beam of RF energy has to be steered to radiate a target. There are many ways of doing this, however, with the advent of solid-state devices, RADAR arrays use electronic beamsteering in what are called phased arrays or electronically steered array RADAR (ESAR) [3]. There are various architectures of ESARs, most notably the active electronically scanned arrays (AESA) where the transmit and receive elements are distributed to each radiating element in an array to overcome RF losses [3].

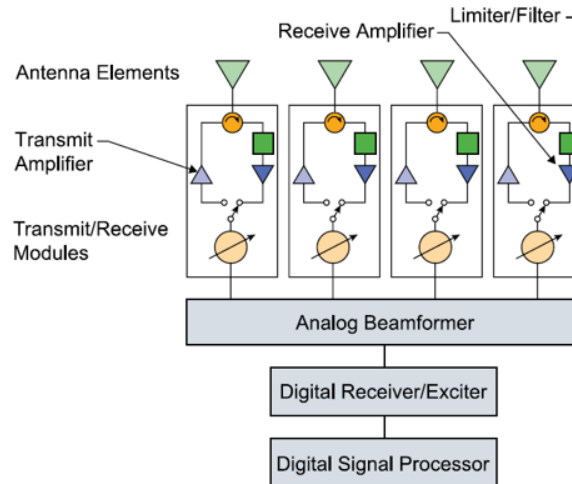
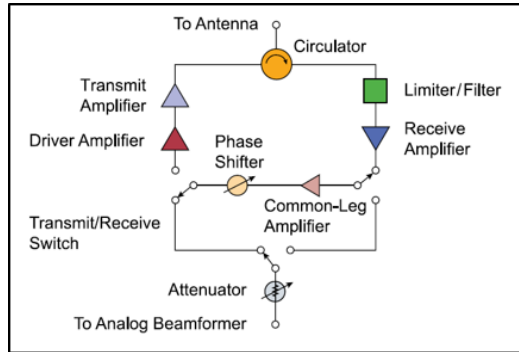
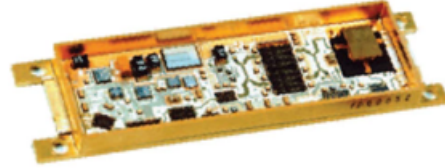


Figure 1: AESA Block Diagram [3]

AESAs use subarray components such as transmit/receive (T/R) modules to distribute active components throughout the array which offers a lower noise figure (NF) and therefore improved sensitivity and longer range [3]. T/R modules also offers a solution if some RF channels fail, the array may maintain acceptable performance with a 5%-10% degradation. Figure 1 shows a block diagram of an AESA while figure 2a shows the block diagram for a T/R module and a photograph of a X-band T/R module from Raytheon in figure 2b.



(a) T/R Module



(b) Active X-band T/R module (used with permission from Raytheon)

Figure 2: T/R module block diagram and photograph [3]

The T/R module is comprised of separate transmit and receive paths with specific components for each in addition to a common-leg path that is used for both transmit and receive. The common-leg path typically contains amplification stages, attenuators, and the phase shifter for beamsteering. Within the transmit path lies an amplifier that drives a larger power amplifier (PA) which is responsible for the transmit power of the RADAR. The receive path contains a limiter or filter that helps protect the low-noise amplifier (LNA) which is responsible for the RADAR's sensitivity. These two paths are joined by a circulator which only allows RF energy to propagate in one direction and is also connected to the antenna element. The RADAR functions by transmitting an RF pulse out through the common-leg and transmit path which then radiates out through the antenna. The RF energy hits a target and returns through the antenna, limiter, and LNA before going through the common-leg again to determine the trajectory of the target.

Most of these components in the T/R module are separate MMICs including the driver amplifier (DA), the PA, limiter, and LNA. The common-leg circuit which again incorporates the common-leg amplifier, attenuators, and phase shifter are typically integrated into a single MMIC by the moniker of a multi-function chip (MFC). Seeing as the MMICs, especially the high power amplifiers, are the largest sources of heat, cooling these components becomes a primary task in RADAR sub-assembly architecture design. However, there are always trade-offs when considering which architecture to deploy such as size, power, thermals, efficiency, cost, and size also referred to as SWAP-C.

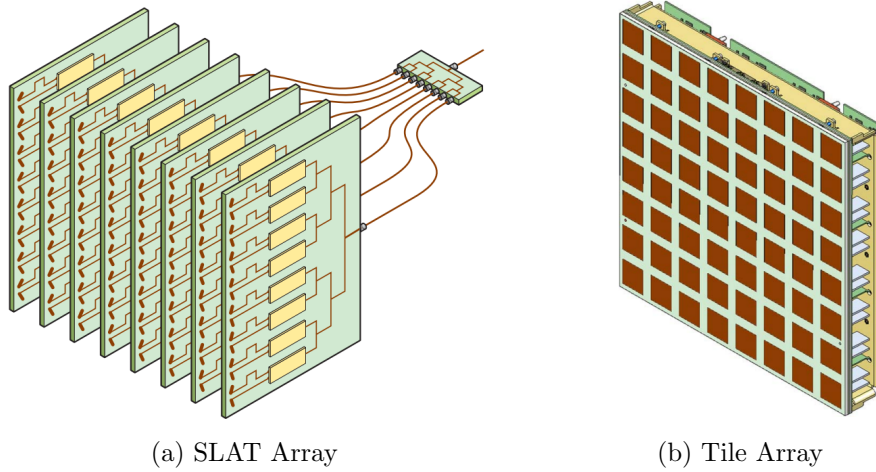


Figure 3: SLAT and Tile Array Configurations [3]

There are two main configurations of an AESA which are the SLAT array and Tile array. SLAT arrays, seen in figure 3a, use multiple PCB boards with T/R modules and antenna elements integrated into it. This can be very costly, however, is an excellent choice for thermal management. Tile arrays, seen in figure 3b, use a single PCB board that connects all of the T/R modules and antenna arrays on top of it. This is very cost effective but typically has poor thermal management as all of the high power components are stacked on top of each other. However, for mmWave frequencies, it is becoming more useful to deploy tile configurations as the element to element spacing between antennas must follow equation 1.

This project will explore certain intricacies within MMIC design that will allow for these tile arrays to be used which could offer to be easier to manufacture as well as reducing the amount of external cabling and connectors used. Until now this tile configuration required vertical chip stacking which provided poor thermal management and chip-to-chip isolation. This project aims to solve this issue by integrating a MMIC vertically using stacked stripline design so that the overall footprint of the chip remains minimal while maintaining maximum RF performance at mmWave frequencies. There are prior efforts to implement a stripline design within MMIC environments, however, not at mmWave frequencies.

2.2 Monolithic Microwave Integrated Circuit (MMIC)

In the past, RADAR's used vacuum tube amplifiers to achieve the required transmit power in order to achieve an effective range. In fact, Raytheon had pioneered these tube amplifiers for commercial use and when the need for high power amplifiers arose during World War II, it would be Raytheon who supplied these to the U.S. Military. These tube amplifiers would remain the primary technology for high power applications as they were relatively small in size for their performance, at the cost of the lifespan of the filament used inside the vacuum tube.

Since the late 1970's, integrating circuits onto a single chip instead of using a hybrid form of chip transistors and printed circuit cards (PCB), has been made possible with the advent of different materials such as GaAs and GaN [4]. MMICs incorporate all the passive and active components into a single substrate which greatly reduces the cost and size of a device. This also allows for fewer transitions from different substrate mediums which inherently will cause signal loss and therefore inefficiencies. Figure 4a is a diagram that shows a MMIC layout and how various components are arranged and Figure 4b is a CAD photograph of a GaAs X-band amplifier MMIC.

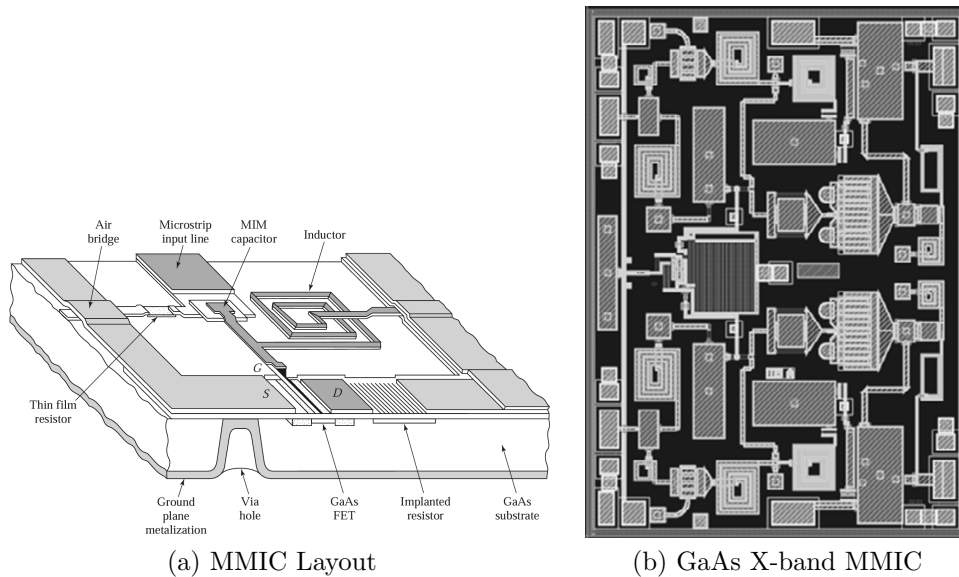


Figure 4: MMIC Layout and Photograph. Photograph courtesy of M. Adlerstein, Raytheon Company, Waltham, Mass.[5]

2.3 Microstrip vs. Stripline

Typically MMICs use either microstrip or coplanar waveguide (CPW) transmission lines. Microstrip refers to a topology where the conductor or signal line is on top of the substrate with a ground plane on the bottom of the substrate. Coplanar waveguide is similar to microstrip where the signal line is on top of the substrate but with the ground plane on the same side as the conductor and separated by some distance. Figure 5a shows the layout of microstrip and how the electric (E) and magnetic (H) field lines propagate. Figure 5b shows the layout of a coplanar waveguide transmission line.

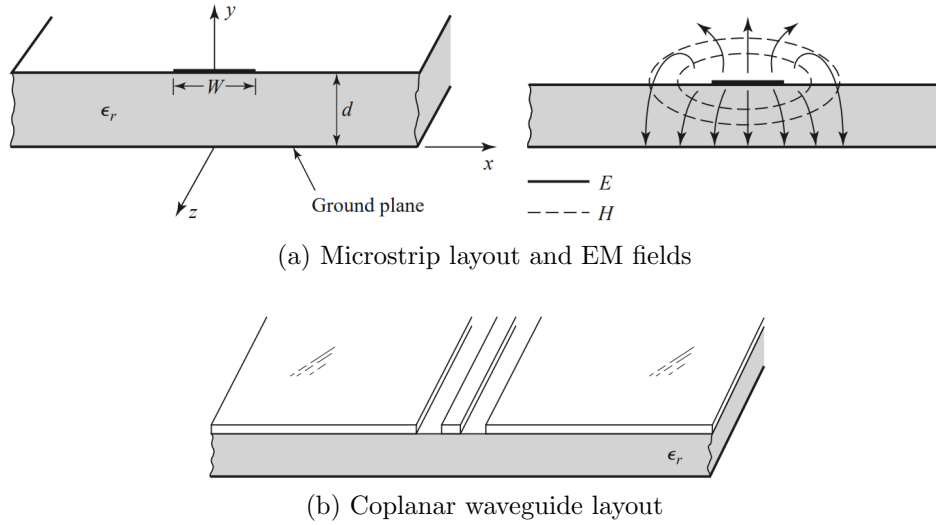


Figure 5: Microstrip and CPW Transmission Lines [4]

These transmission line layouts are typically used since they are easy to manufacture, especially within MMICs as metal and other dielectric layers are deposited on top of the gallium nitride dielectric crystal. Also, the dielectric thickness dictates how wide the transmission lines have to be in order to obtain a certain impedance which then translates into the overall chip size, as indicated by W and d in figure 5a. As shown in figure 5a the electric fields will propagate downwards towards the bottom ground plane in a microstrip line whereas in a CPW line (figure 5b), the electric field will propagate towards the adjacent ground planes surrounding the conductor line. However, the electric field will also propagate upwards, away from the signal line, as marked by the y -axis in figure 5a. With this, great care and consideration

is taken for objects or packaging that encases the MMIC so that there is no external affects that perturb the MMIC's electrical performance. In extreme cases there may be a coupling effect that occurs where the MMIC is not sheilded or isolated enough from the outside environment and undesired results are seen. This is where the benefits of stripline transmission line layout come into affect. Stripline is essentially the same as microstrip but instead of the conductor metal on top of the dielectric and exposed to open air, the conductor metal is now embedded into the dielectric material and another ground plane is placed on top, as seen in figure 6 below.

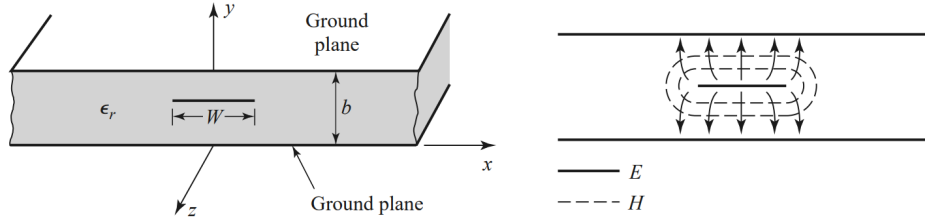


Figure 6: Layout of Stripline Transmission Line [4]

Although stripline topologies have slightly more signal loss, they have far better isolation from the surrounding environment due to the two ground planes and are not typically done in MMICs as it isn't easily implemented due to the manufacturing process.

This project offers a novel approach to using stripline transmission lines instead of microstrip or CPW within a MMIC. A benefit of stripline design that isn't typically feasible within a MMIC is the vertical integration of matching or bias networks that traditionally will take up most of the real estate within a MMIC. This may be acceptable at lower RF frequency bands but at mmWave frequencies and above, spacing between RF radiating components becomes a challenge as the minimum distance follows equation 1 below where λ is the wavelength of the highest desired frequency [6].

$$d_{max} = \frac{\lambda}{2} \quad (1)$$

It is clear to see from equation 1 that as frequency increases and the wavelength decreases, the maximum distance will decrease. The following section will discuss different array architectures that will go into more detail about element spacing and factors behind their use.

2.4 Performance Metrics

RF components are typically referred to networks with a number of ports where a port is the input or output of the device. The number of ports a device is dictated by the application or desired use and can range from one or tens. The most common configuration is a two port device such as an amplifier with an input and output port. Performance for RF components can be categorized in two ways; linear or small-signal and non-linear or large-signal. The following sub-sections will explain each of these in more detail.

2.4.1 Small-Signal (linear)

Linear performance refers to a linear relationship between an input and the response a component has. In the case of an amplifier, a linear response would mean that any change at the input would correlate to a linear change at the output. Something that must be taken into consideration for a device operating at RF or mmWave frequencies is the mismatch of complex impedances (Z) seen in equation 2.

$$Z = R \pm jX_c \quad (2)$$

Complex impedances are composed of a real resistance (R) and imaginary reactance (X_c) where the resistance correlates with the loss of a signal or component and the reactance correlates to the phase of a signal or phase change of a component. The reactance of a component will be inherent of how much capacitance or inductance it has. A mismatch of these complex impedances will cause signals to reflect back toward the generator and is defined as the voltage reflection coefficient (Γ) [4]. The reflection coefficient is derived from equation 3 where Z_L is the complex impedance and Z_0 is the system impedance.

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} \quad (3)$$

These reflections are a product of the components inherent impedance and is known to be a linear phenomenon. An intuitive way to view and manipulate these complex impedances is a Smith chart seen in figure 7. A Smith chart depicts a circular representation of the real and imaginary axis. The real or resistive axis is a horizontal line drawn through the middle of the circle which divides it into top (inductive) and bottom (capacitive) halves,

figure 7. Typically, a Smith chart is normalized to the system impedance which in most cases for RF systems is 50Ω . The center of the Smith chart corresponds to the system impedance and acts as the main reference point to where Γ can be derived from.

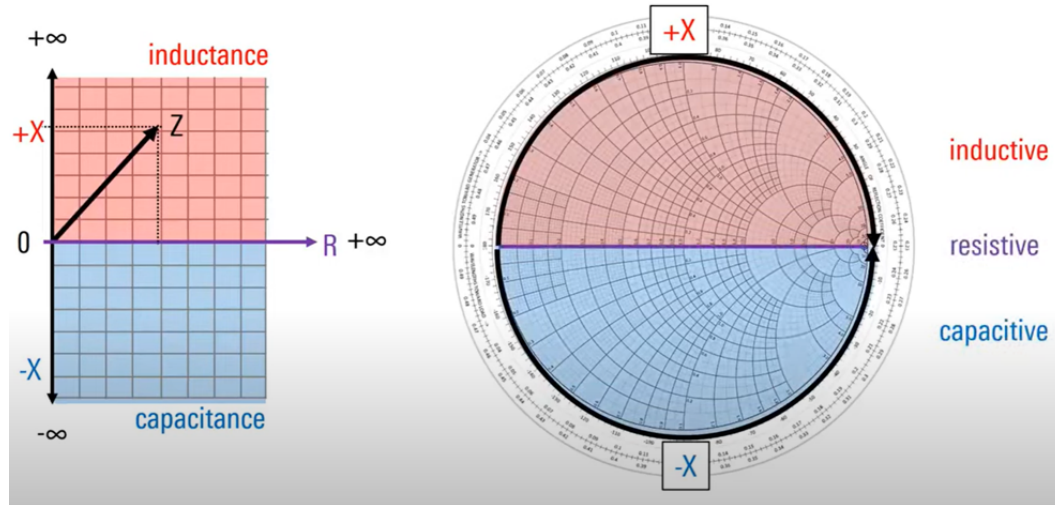


Figure 7: The Smith chart [4]

Often it is convenient to plot the magnitude of Γ or $|\Gamma|$ on the Smith chart which will show a circle in which any load along will produce a mismatch of that Γ . This can be seen in figure 8. Another way many RF engineers choose to look at the reflection between components is the voltage standing wave ratio (VSWR). When a wave is reflected back due to an impedance mismatch there is an overlap of the incident and reflected wave, [4]. VSWR is a ratio which tells the engineer how well the component is matched. A perfect match has a VSWR of 1:1 where a mismatched component may have a VSWR of 1.2:1 or 2:1. Specifications may dictate that a component must be within a certain VSWR which correlates to how well matched it must be to the system impedance. Tied to both $|\Gamma|$ and VSWR is return loss (RL), in dB. Return loss is the logarithmic representation of $|\Gamma|$ and defines the ratio of the available power from the generator delivered to the load. With this mismatch of a load or component there is a power mismatch loss (ML), also in dB. Equations for $|\Gamma|$, VSWR, RL, and ML can be seen in equation 4.

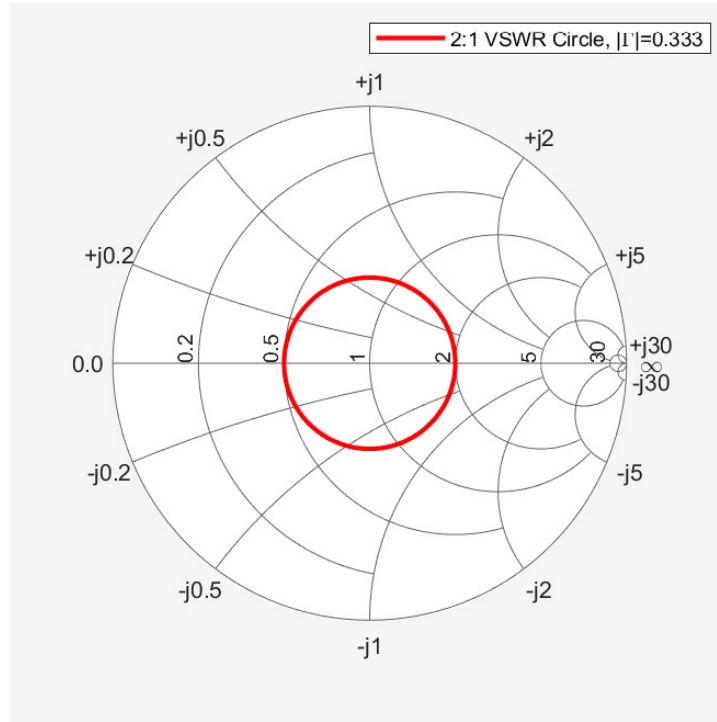


Figure 8: Constant VSWR and $|\Gamma|$ circle plotted on Smith Chart

$$\begin{aligned}
 VSWR &= \frac{1 + |\Gamma|}{1 - |\Gamma|} & RL &= -20 \log |\Gamma| \\
 |\Gamma| &= \frac{VSWR - 1}{VSWR + 1} & RL &= -20 \log \left(\frac{VSWR - 1}{VSWR + 1} \right) \\
 ML &= -10 \log (1 - |\Gamma|^2) & ML &= -10 \log \left(1 - \left[\frac{VSWR - 1}{VSWR + 1} \right]^2 \right)
 \end{aligned} \tag{4}$$

A relationship between the incident, transmitted or forward, and reverse or reflected waveforms going into and reflecting from a device can be quantified in a matrix form as the scattering matrix [4]. The scattering matrix is composed of a number of terms proportional to the number of ports a device has. For an N-port device it will inherently have an S_{NN} sized matrix. Each term in the matrix corresponds with the direction of each forward and reverse waveform. For example, in a two port network like an amplifier, there will be four wave components. These are noted by a_1 , b_1 , a_2 , and b_2 in figure 9 which are waves measured in power (W) or in log form of decibels with reference to 1 mW known as dBm. Scattering parameters or S-parameters are the ratios of these waves as each port is terminated to the system impedance. Typically these ratios are measured in decibels or dB. Equation 5 shows the s-parameters and their ratios of a two port network.

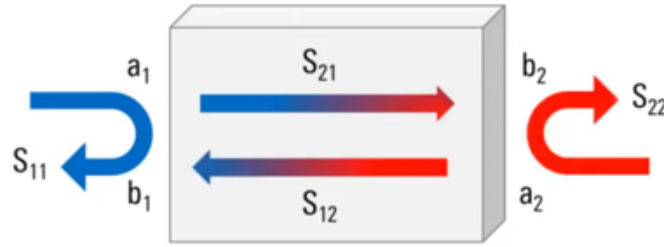


Figure 9: 2-port S-Parameters [7]

$$\begin{aligned}
 S_{11} &= \left. \frac{b_1}{a_1} \right|_{a_2=0} & S_{12} &= \left. \frac{b_1}{a_2} \right|_{a_1=0} & a_1 &= \frac{V_1^+}{\sqrt{Z_1}} & b_1 &= \frac{V_1^-}{\sqrt{Z_1}} \\
 S_{21} &= \left. \frac{b_2}{a_1} \right|_{a_2=0} & S_{22} &= \left. \frac{b_2}{a_2} \right|_{a_1=0} & a_2 &= \frac{V_2^+}{\sqrt{Z_2}} & b_2 &= \frac{V_2^-}{\sqrt{Z_2}}
 \end{aligned} \tag{5}$$

S-parameters are an intuitive way to quantify the linear behavior of an RF component or system. Each S-parameter represents an inherent property of a component and can also explain if a component is passive or active, reciprocal or non-reciprocal. Typically, S_{11} is the input return loss and S_{22} is the output return loss. S_{21} represents the insertion loss for a passive component or the small-signal gain of an active component such as an amplifier. Gain of an amplifier is the ratio of the magnitude of output signal (b_2) to input signal (a_1). S_{12} will represent the same insertion loss if a component is passive and

reciprocal or the reverse isolation of an active amplifier. S-parameters are typically plotted in Cartesian coordinates in dB versus frequency while S_{11} and S_{22} can also be plotted on a Smith chart as they are a direct characterization of the input and output impedance match to the system impedance. Figure 10 shows an example of S-parameters plotted for a Qorvo GaN power amplifier.

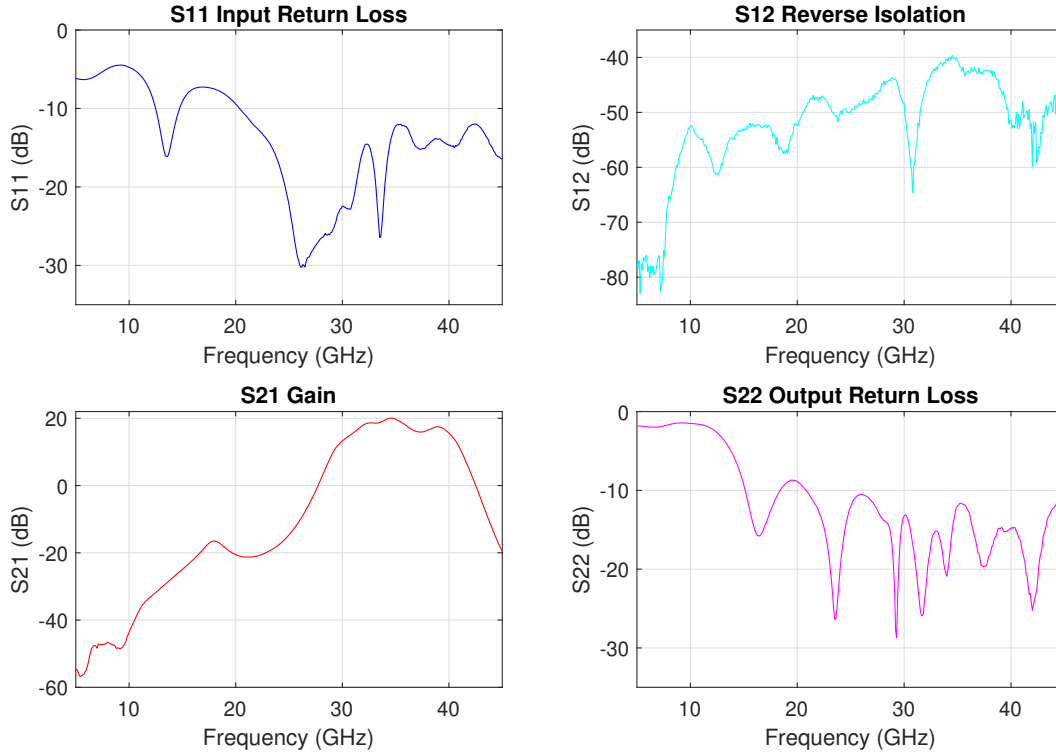


Figure 10: S-Parameters for Qorvo TGA2575 Ka-Band GaN Amplifier [8]

When looking at different s-parameters there are certain features that are desirable and can describe the circuit or component. In the case of the Qorvo amplifier in figure 10 we can see that S_{11} is at least -12dB and S_{22} is at least -10dB in the band of interest of 32GHz. to 38GHz. We can also see that S_{21} or the small-signal gain of the amplifier is at least 15dB in the band of interest. In the case of an amplifier, S_{12} should be as low as possible as to not cause an instability in the amplifier. Stability is a very important concept that will be discussed in the final report for this project.

There are two phenomenon that are relevant to this project that can

be quantified by the small-signal response of a component. Typically these concepts are discussed when a device has more than two ports, i.e. a 3-port or 4-port network. In the case for this project, we can assume that two 2-port networks adjacent to each other in physical space can be considered a 4-port network. Typically a 4-port network is intended to be coupled together, however, at mmWave frequencies 2-port networks must be placed in close proximity to each other where the likelihood of coupling can occur and so it is relevant to consider the isolation between these two adjacent components. The scattering matrix of a 4-port device will have sixteen unique s-parameters, all which correspond to the waves going into and out of each of the four ports. Isolation is the measure of the power delivered to the uncoupled or adjacent port [4]. Coupling refers to the fraction of the input power that is present at the output port of the adjacent network [4]. Figure 11 shows a block diagram of what two adjacent 2-port networks may look like while equation 6 shows the mathematical representation of isolation and coupling within a 4-port network.

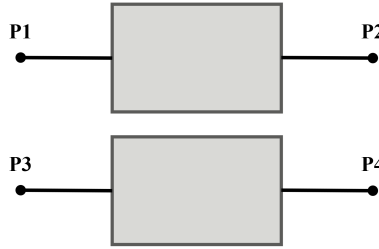


Figure 11: 4-port network block diagram

$$\begin{aligned}
 Isolation(I) &= 10\log\left(\frac{P_1}{P_3}\right) = 20\log |S_{13}| \\
 Coupling(C) &= 10\log\left(\frac{P_4}{P_1}\right) = 20\log |S_{41}|
 \end{aligned}
 \tag{6}$$

It is worth noting that s-parameters are still complex values, meaning they have both magnitude and phase. While most small-signal performance is presented as the magnitude of s-parameters, as seen in figure 10, there are some components that consider the phase of these signals. Components such as phase shifters, dividers, and couplers all use the phase of these signals to perform their designed functions.

2.4.2 Large-Signal (non-linear)

Now, due to the non-linear nature of transistors which are used in most amplifiers, a non-linear response can be seen in RF amplifiers where they will reach a point called saturation. More about transistor physics will be explained in later reports, the key take-away is that any active component that uses transistors, will inherently experience non-linear behavior. This behavior will not be present in small-signal analysis, therefore, analyzing an active circuit requires what is known as large-signal analysis. Large-signal analysis (LS) is the exercising of component behavior with the prescience of DC or voltages and currents where small-signal did not. With voltages applied, device complex impedance may change, and the response to an input signal will also change compared to that of small-signal. With an amplifier, the s-parameter S_{21} will represent the linear gain of the device, keeping in mind that gain is the ratio of output to input signal.

In large-signal analysis, the input signal power is increased so much that the amplifier gain will begin to compress [4]. Compression is the non-linear artifact of transistors within the amplifier and is something that can be optimized in the design process. An amplifier's compression point is where the linear gain will start to decrease until the power output reaches a saturation point, where any change in input signal power will not proportional change the output signal power. Figure 12 shows the LS gain and compression of an amplifier.

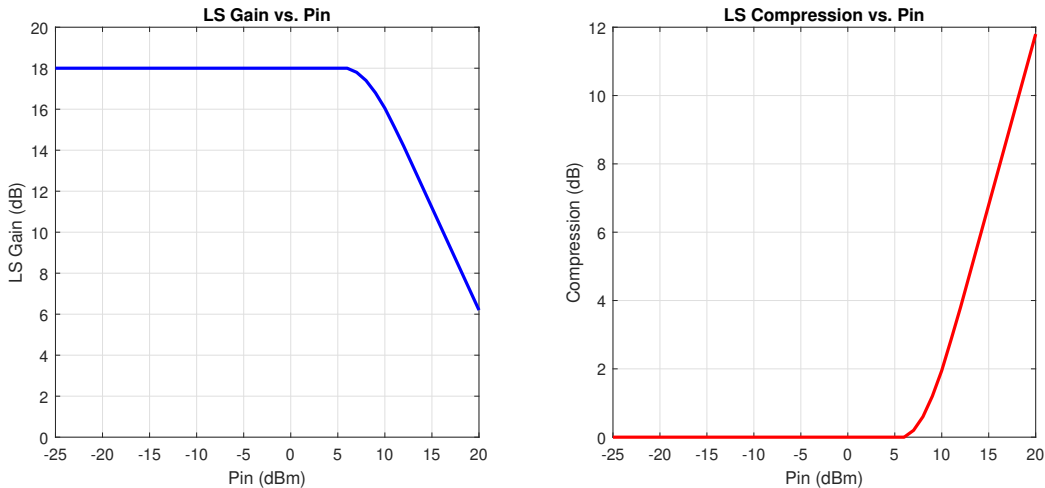


Figure 12: Amplifier LS Gain and Compression vs. Pin

Typically, the output signal power, measured in W or dBm, where the linear gain of an amplifier drops by 1dB is known as the 1dB compression point (P_{1dB}). Tied to P_{1dB} is the measure of linearity, known as the third-order intercept point (TOI) or 3^{rd} order intercept point (IP3), which is measured in W or dBm. To find this, the slope of the linear gain must be scaled by 3 and 5 to find the 3^{rd} and 5^{th} harmonic spectral output of the amplifier. Where the curves of the linear or small-signal gain, 3^{rd} harmonic gain and 5^{th} harmonic gain intersect will be the third-order-intercept. There are two components to IP3 which are input 3^{rd} order intercept point (IIP3) and output 3^{rd} order intercept point (OIP3), both of which are linearity figures-of-merit. The higher IIP3 is, the more linear the device is.

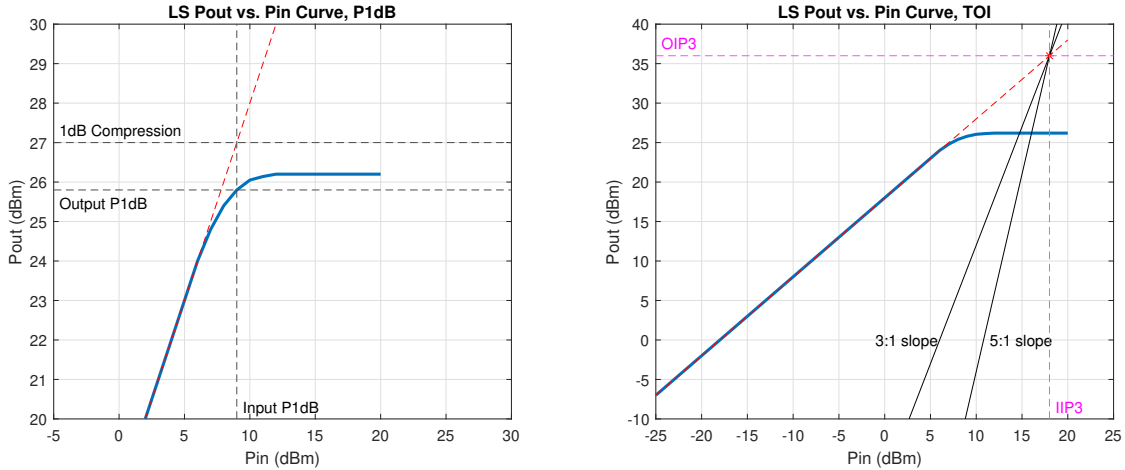


Figure 13: Pout vs. Pin Amplifier Compression Curve

Figure 13 shows a plot of output power versus input power for a typical amplifier. In figure 13 the point at which the theoretical linear gain compresses or drops by 1dB is shown and the input and output power values for the P1dB point can be interpolated from the graph on the left. In figure 13 the IP3 point is shown where the gain slopes of the fundamental, 3^{rd} harmonic and 5^{th} harmonic intersect, and the OIP3 and IIP3 points can be interpolated from the graph on the right.

2.5 Technology Implementation and Prior Art

Finally, prior art or prior work on similar technology by other entities will be discussed. This serves only as evidence that this project is viable from a technology standpoint. Two aspects will be addressed here; the use of benzocyclobutene (BCB) in MMICs and other integrated circuits, and a similar project conducted by another prime defense contractor, Northrop Grumman.

2.5.1 Benzocyclobutene (BCB)

Benzocyclobutene (BCB) is a synthetic polymer developed by Dow Chemical Company. This material is commonly used in modern microelectronics for protecting passive components while having a low dielectric constant, low RF loss, and resistivity to high DC voltages and currents [9]. Companies that manufacture multi-layer monolithic integrated circuits will use BCB for inter-layer dielectric, which is relevant to this project's objectives for making a stripline MMIC with multiple layers of dielectric material [10]. BCB also offers good thermal, chemical and mechanical stability while being compatible with standard integrated circuit wafer processing [11]. Raytheon already owns a recipe for implementing BCB into its MMIC wafers fabrication and has had good success with RF performance and wafer yield which will contribute to the cost analysis and ROI for this project. If Raytheon can leverage its existing BCB capabilities, the use of it to develop multi-layered dielectric MMIC designs for mmWave frequency applications while keeping the cost of average assets down will add to the value of this project to the company.

2.5.2 Prior Work - Northrop Grumman

Northrop Grumman, a competing prime contractor, has developed a GaAs MMIC using multiple layers of BCB to form a stripline layout of a phase shifter for X-band frequencies. Figure 14 shows their process stackup of each BCB and metal layer on top of a GaAs substrate. These layers of BCB and metal form the proposed stripline architecture that will be an objective of this project. Figure 15 shows how different layers layout various components of the X-band phase shifter. The paper Northrop Grumman published for this work boasts how the size of the RF component, phase shifter, reduced by 70% while maintaining the main functionality of the phase shifter [12].

[13]

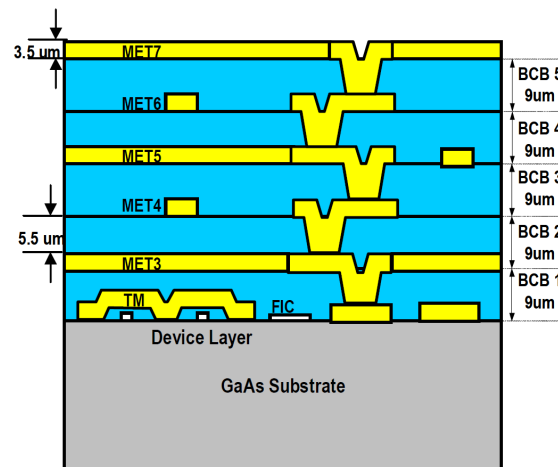


Figure 14: Process stack-up showing each BCB and metal on GaAs substrate [12]

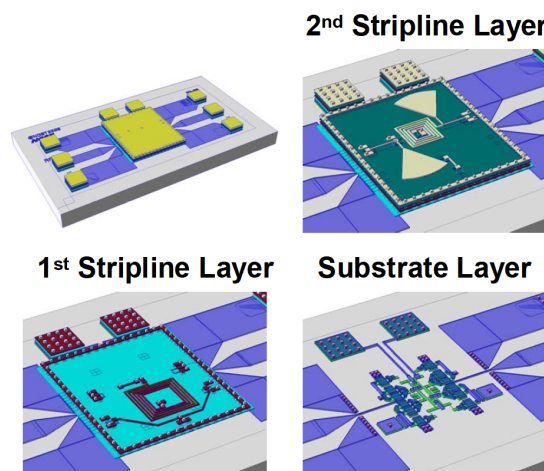


Figure 15: 3D rendering of phase shifter MMIC showing circuit elements on each layer [12]

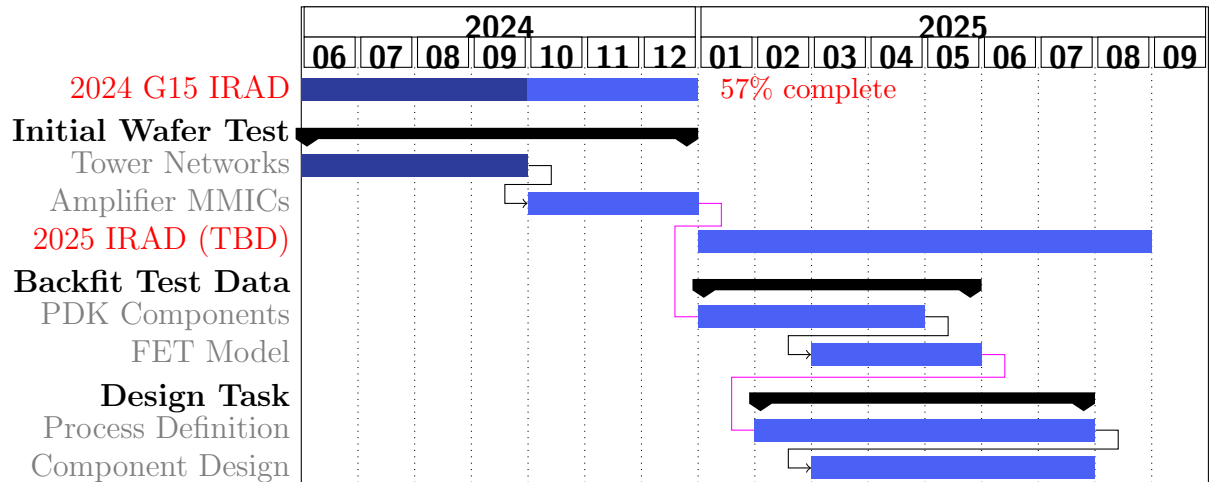
3 Project Plan

3.1 Description of Tasks

Key deliverables for this project may include the following:

1. Backfit devices and process design kit (PDK) components using initial wafer data
2. Investigate the use of BCB for topside layer dielectric
3. Investigate the thinning of III-V substrate and replace with a higher thermal conductive material
4. Contribute to a manufacturable stripline wafer process
5. A finalized design and layout of a stripline MMIC

3.2 Time Plan - Gantt Chart



Above is the proposed Gantt Chart for this project. The timeline for 2024 has been set until the end of the year and will be funded under an internally funded research effort (IRAD). Funding for next year has still yet to be determined and is called out as a risk in the section 3.4.

3.3 Market Value

3.3.1 Value Proposition

As of right now, the wafer foundry at Raytheon in Andover, MA primarily produces microstrip and CPW MMICs. This project proposes to use different topologies such as stripline or inverted microstrip for various high frequency applications. The value in pursuing this effort lies in the fact that it could potentially change the way MMICs are produced, which has more or less been the same for decades. Components would be able to be either stacked on top of each other or placed very close to each other due to the high isolating properties of these unique topologies. In addition to this, the new topologies could allow MMICs to be thermally managed more effectively as they are typically the focal point for many mechanical engineering teams who deal with the thermal management of RADAR systems.

This provides high market value to Raytheon as it would increase our capabilities, especially within the wafer foundry, to produce very reliable, high frequency circuits and integrate them into stacked arrays as well as providing a cheaper solution for the cooling systems in larger arrays. This would broaden Raytheon's catalog of potential products it could offer to the Department of Defense for military applications but could also be a pivot point for existing designs to transition into.

3.3.2 Return on Investment (ROI)

Return on investment or ROI is a ratio, equation 7 below, that indicates how well assets of a company are used in order to produce revenue or returns [14]. This ratio is a key financial indicator if an investment is worth spending time and effort on.

$$ROI = \frac{\text{Net income}}{\text{Average assets}} \quad (7)$$

For this project the ROI would come from Raytheon being able to use this new technology as a selling point to win different contracts with either of the military branches through the DoD. Currently this project is of interest for two different U.S. Government labs which have been involved in various stages including the concept and measurement verification. If new contracts are won based on this technology then the net income would increase as sales increase. Average assets in equation 7 refers to the average value of the assets

a company owns. Since this technology is already leveraging existing tools and operations within Raytheon and the AMS department, there wouldn't be the need for a large investment into this technology in order to develop it. A key point for this project is to investigate the ability of Raytheon's foundry to integrate this technology with existing tools and methods. Therefore, if Raytheon can leverage the technology they already have in order to develop a new product that would produce a net income, the return on investment would increase.

3.4 Risks and Dependencies

3.4.1 Technology Risk

There is an inherent risk in implementing any new technology in that it may not work. Currently, test structures for calibration have already been characterized and full amplifiers are to follow, see Gantt Chart in section 3.2. Initial measurements for the calibration test structures looks promising which indicates that the amplifier measurements may follow but is not a certainty. This being said there is a risk associated with the development of this new process technology in that it may not meet the performance of existing robust technology Raytheon already implements.

3.4.2 Financial Risk

As mentioned in section 3.2, the completion of this project relies on internal research funding from Raytheon. This funding typically spans a calendar year which is then re-assessed at the beginning of each year. This means that the funding that is available in 2024 may not be available in 2025. Funding for the technical deliverables of this project are dependent on Raytheon's internal research funding allocation, despite the interest in U.S. Gov. labs. Future efforts will be dictated based on this funding and changes to the project deliverables will scale accordingly. However, given this project has had initial interest from both the U.S. Gov. and Raytheon, funding for 2025 may come from either Raytheon's internal research or corporate research funds.

4 Personal Impact and Leadership

4.1 Team Leadership

Leadership is an essential part of this project, not only in the specific project requirements but also in how the AMS department is run within Raytheon. There are various roles that portray varying levels of leadership through a project such as this one. Not one single engineer works in isolation as interdisciplinary team members must work together in order to complete their specific tasks on time and on target. Figure 16 shows a typical team in the AMS department at Raytheon along with their connections. This figure 16 is from the perspective of the MMIC designer. Of course the connections between other discipline engineers will change from other role's perspectives, however, the key personal is typically the same.

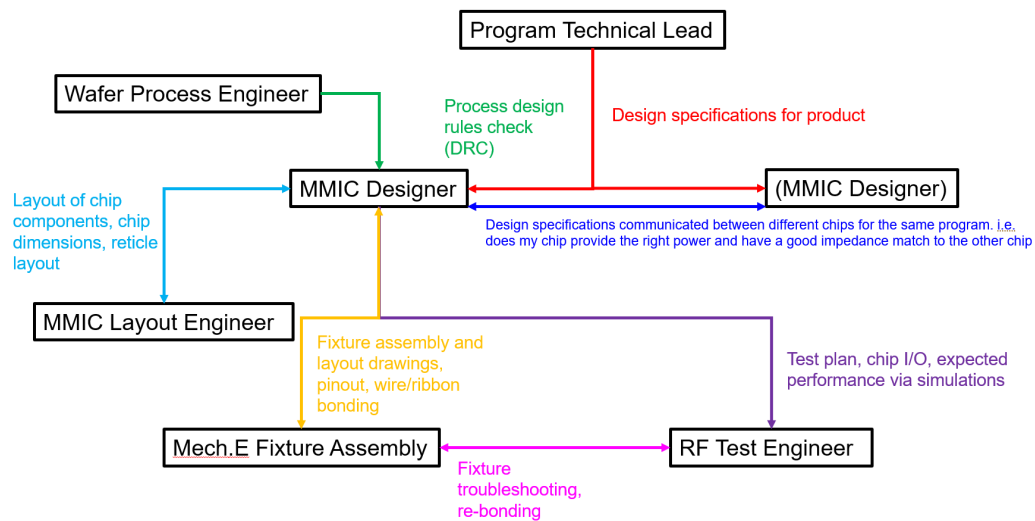


Figure 16: Typical AMS Team at Raytheon

In addition to this, a MMIC designer who is tasked to design a microchip to certain specifications is also required to hold design reviews throughout the design process. These design reviews provide a snapshot in time of where the design is and comes with feedback from multiple sources including but not limited to the people listed in figure 16 as well as independent reviewers who are not part of that specific program or project. During these design reviews a MMIC designer must be able to convey his or her work and findings

in a clear and concise manner so that the key stakeholders of the project are well informed.

4.1.1 Design Reviews

Design reviews in the AMS department play a crucial part in a MMIC design being released into production by the foundry. Although their contents are mostly technical, they serve as an opportunity for the candidate to develop leadership skills. There are five major design reviews for MMICs, however, not all designs go through each review as some designs are just transferring from a development to a production mask. As each design review is held, the level of technical depth increases, along with the weight of each design features' justification as to why it should be made. These justifications are made to senior technical staff who act as peer-reviewers, who's task during the design review is to formulate questions as to why or how each feature in the design was developed or calculated. Championing these design reviews gives the candidate a chance to stake their reputation and trust within the eyes of their peers and mentors.

The following are the 5 design reviews for the MMIC Design section of the AMS department:

1. **Advanced Design Review (ADR):** Design kickoff, intended on understanding the required specifications of the design and any legacy designs that may have pieces or structures that could be used in order to achieve the required specifications. Topics such as frequency band, small-signal and large-signal parameters, chip size, and process technology (GaAs or GaN) are discussed. This design review is more of an open conversation between the design team and the technical director.
2. **Preliminary Design Review (PDR):** Initial design and layout development, legacy components are simulated and backfit to any measurement data that is available. Determine what changes need to be made in order to achieve the required specifications. If any risks exist, they must be documented. ADR action items must be completed and signed off by management. This is the first technically deep design review where the MMIC designer(s) must justify any decisions made for the chip. The onus is on the designer to produce a clear and concise design review package and present to management, senior technical staff, and any internal or external customers.

3. **Critical Design Review (CDR):** The design review with the most technical depth, where all design features, simulations, and calculations must be presented and justified. In some ways this is the most important design review for the designer. The designer must be able to present on every aspect of the chip including its RF and DC performance, the physical size and layout, and reliability according to the inherent process variation (6σ). If there are multiple chip variations then each must be presented, clearly showing the performance differences between each chip. In some cases where the chip has multiple components, then multiple designers will present on each portion they contributed to. PDR action items must be completed and signed off by management. Personal from each major section in the department must be present in addition to at least two independent reviewers.
4. **Chip Release Review (CRR):** This design review is intended to prepare the design for release to the foundry. Here the design's final layout is shown along with the final chip dimensions, typically in units of millimeters (mm) or thousandths of an inch (mils). Test pads for the chip are also laid out which are where connections will be made to or from other components by wire or ribbon bonds. One of the main purposes of CRR is to review the DC and RF test plans for the design where respective teams will confirm how the chip will be tested once out of the foundry. While this design review is still the MMIC designer's responsibility, there is more collaboration with the layout engineer to make sure the design passes all design rule checks (DRC) as well as making sure the layout matches the design schematic (LVS). This is the final design review the MMIC designer is primarily responsible for.
5. **Fab Release Review (FRR):** This design review's purpose is to ensure that all teams from the foundry are prepared for the design to be fabricated. Final layout, chip size, and reticle are discussed. The process and layout engineers discuss any unique challenges that may arise with the design. Once agreed upon, the mask is ordered for the design. The mask is a set of lenses for each layer that will be deposited onto the wafer and is necessary for the fabrication of all MMICs. Once FRR is complete the design is uploaded to a central design repository and production can start. The design is complete.

4.1.2 Leading from the middle

Leading from the middle is defined as being someone who is not in a prominent position or at the top of an organization and has a 360-degree responsibility to influence peers, customers, and direct managers [15]. Graphically this can be seen in figure 16 and can be deducted from the design review descriptions in section 4.1.1. Leading from the middle will be the primary leadership conduit for this project as the candidate has to influence senior technical staff who are funding the project, colleges who may be aiding with specific tasks for the project, as well as internal customers within Raytheon to propose the investment and adaptation of this technology.

Now, understanding that this project as a whole has a significant impact on how Raytheon designs and produces integrated circuits or MMICs, situational blindness is inevitable. *Situation blindness* is the notion that the candidate will not fully see what is happening at the top level of the organization or laterally within other roles [15]. The onus is on the candidate to ensure that all of the relevant information is being conveyed and nothing within the project's scope is missed. The candidate's role for this project is to investigate it's viability from a MMIC designer's perspective, however, it will be important for the candidate to seek out higher level details that may influence the trajectory of the project.

Within the concept of leading from the middle is the notion of tearing. *Tearing* is the idea that as the candidate's role is in the middle, they will experience influence and pull from upper level leadership but also from adjacent or lower level roles [15]. Specifically for this project, tearing may occur in a non-traditional way where demands or needs from other disciplines may influence how the candidate approaches the development of the technology. For example, the perspective from a design engineer may consider the electrical performance of the technology while other engineers in the mechanical or industrial engineering disciplines may consider the thermal effects or manufacturability of the technology. It is pertinent that the candidate takes all of these perspectives into consideration to ensure that the technology is viable from an RF perspective but also conforms to Raytheon's standards for production and manufacturing.

Again, the onus is on the candidate to manage their time, energy, and focus to ensure that all of these aspects are well managed. For the candidate, there is a standard that will drive their actions, keeping in mind the key stakeholders as the driving factor behind their actions.

4.2 Personal Impact

While this project offers a level of technical development, a main purpose of the *Gordon Institute for Engineering Leadership* is to develop leadership in it's candidates. This comes in two forms; team leadership and how the candidate leads within a team, and personal leadership which is how the candidate holds himself. The GIEL program at Northeastern University defines 14 leadership qualities that form the fabric of team and personal leadership development. Candidates were asked to assess their initial competency in these leadership qualities in the form of a spider chart. The spider chart represents the personal evaluation of each leadership quality, quantitatively scoring each with a value ranging from 1 to 10. While this is an initial assessment, the candidate realizes that these values may change depending on their development day-to-day, week-to-week, or depending on the circumstances the candidate is faced with. This personal development and assessment may not only pertain to the candidate's challenge project, but also their primary role within their employer, as a student at Northeastern University, or in life itself. Figure 17 depicts the initial assessment and proposed growth from this project.

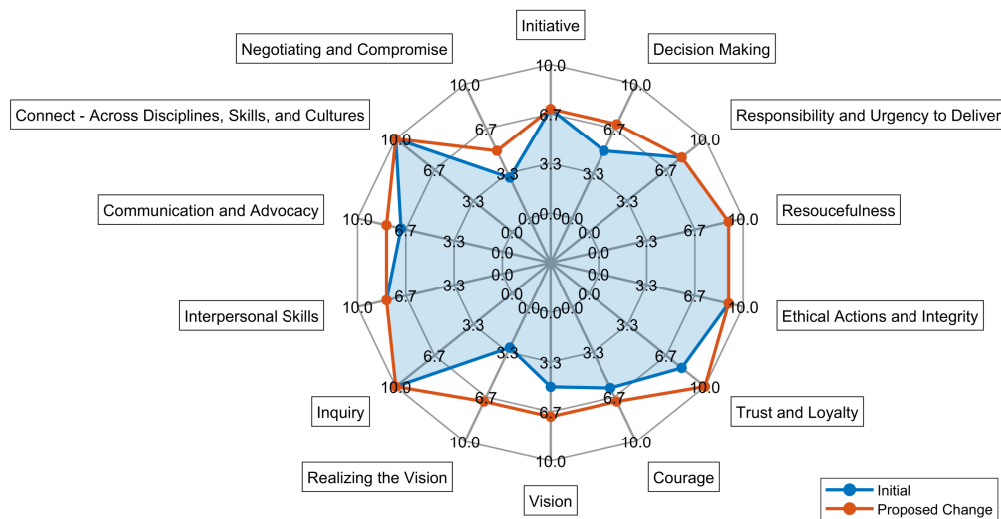


Figure 17: Candidate Spider Chart

4.3 Mentorship

Mentorship is a crucial part of this project, not only technically but in terms of the impact to the sponsoring company and leadership development of the candidate. While there are clearly defined mentors for this project, unofficially the candidate has mentors within their organization that act as guides toward technical competency and personal development. Specifically for this candidate, the AMS department's staff has multiple *Technical Fellows* who have been recognized by the organization as someone who has the highest level of technical knowledge and understanding while also providing a meaningful impact to the organization. The candidate leverages their relationships with these technical fellows in order to fully understand the underlying concepts but also the impact their efforts make. Specifically for this project there are three mentors, each with specific roles as to ensure the candidate's project meets the requirements of the *Gordon Institute of Engineering Leadership*. The GIEL mentors are as follows:

- **Gordon Mentor:** Paul Head, *Senior Technical Fellow at RTX - Raytheon*

Paul started at Raytheon in the former AMS department, RF Components, as a MMIC designer. Paul is now a Director of the Advanced Technology business unit within Raytheon and has been with the company since 2005. Paul Head received his B.S. degree in Electrical Engineering at Northeastern University in 2005 and his M.S. degree from Northeastern University in 2007. While pursuing his M.S. degree, Paul was part of cohort 1 in the Gordon Institute of Engineering Leadership.

- **Industry Sponsor:** Nicholas Kolas Ph.D, *Principal Technical Fellow at RTX - Raytheon*

Research focus in GaN, GaAs, and SiC semiconductor technology, RF and mmWave integrated circuits, power amplifier design, low-noise amplifier design, pseudomorphic high-electron-mobility transistor technology, high-power and high-efficiency MMIC design, etc. Nick Kolas received his B.S. degree, M.S. degree, and Ph.D in Electrical Engineering from Cornell University in 1990, 1993, and 1996 respectively. Nick served as the 2012 President of the IEEE Microwave Theory & Techniques Society. Served as advisor to Paul Head for his GIEL Challenge Project for Cohort 1.

- **Faculty Advisor:** Marvin Onabajo Ph.D, *Professor in the Electrical and Computer Engineering Dept. at Northeastern University*

Research focus in design of analog, radio frequency, and mixed-signal integrated circuits; built-in test and calibration techniques for systems-on-a-chip; on-chip temperature sensors for thermal monitoring and built-in testing; integrated circuits for medical applications. Prof. Onabajo received a B.S. degree (summa cum laude) in Electrical Engineering from The University of Texas at Arlington in 2003 as well as the M.S. and Ph.D. degrees in Electrical Engineering from Texas A&M University in 2007 and 2011, respectively.

5 Summary

The purpose of this proposal is to demonstrate the need for vertically integrated monolithic microwave integrated circuits for wide-band phased array applications. Throughout this proposal a technical definition of key performance metrics, novel design features, and relevance to the application of mentioned components has been clearly defined. A project plan in terms of time, feasibility, and risk management has been developed and is underway. The impact of this project, both to the sponsoring company, Raytheon, and personally has been carefully described in detail. Both the technical aspect and leadership component of the Gordon Institute of Engineering Leadership Challenge Project have been met and will continue to develop as the project progresses.

Success of this project relies on the technical development from the perspective of an electrical engineer, in the role of a MMIC designer. Design results will be presented in detail as to lay out the foundation for Raytheon to implement this into practice. Leadership development will take place in the form of mentorship, mid-level role leadership in the form of leading from the middle, and team leadership within the Advanced Microelectronic Solutions department at Raytheon. This will all be done with the key stakeholders in mind, and while holding the mission statement of protecting the personal of the U.S. Military who protect and serve our nation.

References

- [1] T. Martin and A. Roque, “Ukraine FM calls out Patriot delays, leads with allies to ‘finally deliver’,” *Breaking Defense*, 2024. [Online]. Available: <https://breakingdefense.com/2024/08/ukraine-fm-calls-out-patriot-delays-pleads-with-allies-to-finally-deliver/>
- [2] J. Judson, “How Patriot proved itself in Ukraine and secured a fresh future,” *Defense News*, 2024. [Online]. Available: <https://www.defensenews.com/land/2024/04/09/how-patriot-proved-itself-in-ukraine-and-secured-a-fresh-future/>
- [3] J. S. Herd and M. D. Conway, “The Evolution to Modern Phased Array Architectures,” *Proceedings of the IEEE*, vol. 104, no. 3, pp. 519–529, 2016.
- [4] D. M. Pozar, *Microwave Engineering*, 4th ed. John Wiley and Sons Inc., 2012.
- [5] ———, *Microwave Engineering*, 4th ed. John Wiley and Sons Inc., 2012, ch. 11: Active RF and Microwave Devices.
- [6] C. A. Balanis, *Antenna Theory Analysis and Design*. John Wiley and Sons Inc., 2016.
- [7] Rohde and Schwarz, “Fundamentals of Vector Network Analysis,” *products*, 2024. [Online]. Available: <https://www.rohde-schwarz.com/us/products/test-and-measurement/analyzers/network-analyzers/fundamentals-of-vector-network-analysis>
- [8] Qorvo Inc., “TGA2575,” *products*, 2024. [Online]. Available: <https://www.qorvo.com/products/p/TGA2575/>
- [9] D. Wu, X. Yu, Y. Peng, and M. Feng, “Benzocyclobutene (bcb) process development and optimization for high-speed gaas vcsels and photodetectors,” in *CS MANTECH 2021 - 2021 International Conference on Compound Semiconductor Manufacturing Technology, Digest of Papers*, ser. CS MANTECH 2021 - 2021 International Conference on Compound Semiconductor Manufacturing Technology, Digest of Papers. CS Mantech, 2021, pp. 185–188.

- [10] C. Wong and R. Tummala, "Polymers for electronic packaging in the 21st century," in *Applied Polymer Science: 21st Century*, C. D. Craver and C. E. Carraher, Eds. Oxford: Pergamon, 2000, pp. 659–676. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/B9780080434179500350>
- [11] L. Li, K. Deng, W. Xia, N. Zhu, S. Li, W. Su, and W. Zhang, "A convenient wafer level bonding based on unpatterned bcb," in *The 9th IEEE International Conference on Nano/Micro Engineered and Molecular Systems (NEMS)*, 2014, pp. 502–506.
- [12] D. S. Farkas, J. Uyeda, J. Wang, W. B. Luo, R. Elmadjian, D. Eaves, K. Luo, R. Lai, M. Barsky, M. Wojtowicz, and A. Oki, "Demonstration of a 3-d gaas hemt phase shifter mmic utilizing a five layer bcb process with seven metal layers," in *2007 IEEE Compound Semiconductor Integrated Circuits Symposium*, 2007, pp. 1–3.
- [13] A. Huynh, M. Karlsson, and S. Gong, "Mixed-mode s-parameters and conversion techniques," in *Advanced Microwave Circuits and Systems*, V. Zhurbenko, Ed. Rijeka: IntechOpen, 2010, ch. 1. [Online]. Available: <https://doi.org/10.5772/8419>
- [14] C. K. Drummond, *Financial Decision-Making for Engineers*. Yale University Press, 2018.
- [15] Northeastern University - GIEL, "Leading From The Middle," *GIEL Leadership Fundamentals*, 2016.